Keshav Sundararaman

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EDUCATION

University of Texas at Arlington — M.S. Computer Science — Fall 2025 – Summer 2027

- GPA: 4.0 / 4.0 — Coursework: Data Engineering for Business Applications, Machine Learning, Pattern Recognition, Database Systems

Amrita Vishwa Vidyapeetham — B.Tech Computer Science — Oct 2020 – June 2024

- CGPA: 8.06 / 10 — Thesis: Ensemble ML Pipeline Generation (DSGE)

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Data Analysis:** Pandas, NumPy, Matplotlib, Excel, exploratory data analysis
- **Cloud:** Google Cloud Platform, BigQuery (in progress, via Google Cloud Data Analytics coursework)
- **Machine Learning:** scikit-learn (classification, feature engineering, model evaluation), PyTorch (foundational)
- **Tools:** Git, Linux

EXPERIENCE

Graduate Research Assistant, UTARI — UT Arlington — Jan 2026 – May 2026

- Built a data pipeline that synced and processed RGB and depth sensor streams from ROS bag files, running detection (YOLO-World), segmentation (MobileSAM) and depth projection stages into persistent 3D object maps for robot perception.
- Identified detection speed as the main bottleneck (GroundingDINO alone ran under 10 fps); switched the detection stage to YOLO-World, bringing throughput to roughly 25 fps average.
- Designed a two-pass pipeline architecture separating geometry-only map building from semantic data population, with a configurable CLI, ROS2 Nav2 export and multiple structured output formats (floor-plan PNGs, annotated video, JSON summaries).

Research Assistant, Industry–Academia Project (Sony), India — May 2023 – Dec 2023

- Pulled embedded hardware sensor data over network interfaces and processed it into structured datasets.
- Used the datasets to analyze food drying behavior and reported findings to project stakeholders.

PROJECTS

Abnormal Move Prediction in S&P 500 Stocks — GitHub — May 2026

- Built an end-to-end ML pipeline in Python to predict abnormal next-day price moves across 505 S&P 500 stocks (619K+ daily OHLCV rows, 5 years), engineering 12 no-lookahead features from price and volume history.
- Used strict time-based train/validation/test splits to prevent data leakage; tuned hyperparameters on validation F1 across four model families (L1/L2 logistic regression, linear SVM, random forest).
- Random Forest performed best (test AUC 0.72, F1 0.31 at a 2% move threshold); used feature importance and L1 sparsification to show volatility and price-range features carried the most signal.

Thesis: Generating Ensemble ML Pipelines using DSGE — Aug 2023 – May 2024

- Designed grammatical representations for ensemble ML pipelines using Dynamic Structured Grammatical Evolution, then built Python systems to generate, train and evaluate pipelines automatically across 50+ datasets.
- Ran controlled experiments comparing convergence to strong-performing pipeline configurations against baseline search strategies.
- Validated results with ablation studies isolating the contribution of each pipeline component.

CERTIFICATIONS & COURSES

- Machine Learning Specialization — Stanford University & DeepLearning.AI (Coursera)
- Google Cloud Data Analytics Certificate — in progress
- INSY 5337: Data Engineering for Business Applications, UT Arlington — in progress (dimensional modeling, ETL/ELT pipeline design, Apache Spark & Databricks)